Clustering and UMAP using scRNA-seq data for larvae

Load data

Load the larval data from Pandey et al. (2018).

```
library("data.table")
```

```
##
## Attaching package: 'data.table'
```

```
## The following object is masked from 'package:base':
##
##     %notin%
```

```
read_scs = function(file){
  dt = fread(file)
  dtmt = as.matrix(dt[, -1])
rownames(dtmt) = dt[, V1]
colnames(dtmt) = colnames(dt)[-1]
return(dtmt)

}

create_scs = function(counts,metadata=NULL){
  # Create list and name list elements
  list_out <- vector(mode = "list", length = 3)
  names(list_out) <- c("counts", "initial_dim", "metadata")

  # counts data
  list_out$counts <- counts

  #  `initial_dim`: a vector containing the initial number of genes and cells contained in the raw coun
  dimdata <- dim(counts)
  names(dimdata)<- c("gene","cell")
  list_out$initial_dim <- dimdata

  # `metadata`: dataframe of metadata
  list_out$metadata <- metadata


  # output
  return(list_out)


}
```

```r
qc_scs = function(scs, min_cells=0, min_features=0, mt_genes=NULL, max_mt=100){

# get genes less than min_cells
#   * `min_cells`: remove genes that were detected in less than `min_cells` cells; default value = 0
# rowSums(obj[['counts']] != 0 counts non-zero elements of the row, counting cells that contain that ge

to_rm_genes <- names(which(rowSums(scs[['counts']] != 0 ) < min_cells))

# get cell IDs with less than 'min_features' detected genes
# * `min_features`: remove cells that contain less than `min_features` detected genes; default value =
# colSums(obj[['counts']] != 0 counts non-zero elements of the col, counting number of non-zero count g

to_rm_cells <- names(which(colSums(scs[['counts']] != 0 ) < min_features))


# if mt_gene is provided calculate percentage of mt gene and get cell IDs above max_mt
if (!is.null(mt_genes)){
  tmpscs = percentage_features(scs, mt,"percent_mt")

  to_rm_cells_mt <- rownames(tmpscs$metadata)[which(tmpscs$metadata > max_mt )]

} else {

  tmpscs <- scs
}


# remove genes and cells

tmpscs = rm_cells(tmpscs, to_rm_cells)
tmpscs = rm_genes(tmpscs, to_rm_genes)
tmpscs = rm_cells(tmpscs, to_rm_cells_mt)

return(tmpscs)


}


percentage_features = function(scs, features, name){

  # get the count matrix
  ctmat <- scs$counts

  # get genes containing features expression

  ftarr <- character()
  for (i in 1:length(features)){
    ft = grep(features[i], rownames(ctmat), value = TRUE)
    ftarr <- append(ftarr, ft)
  }

  # feature counts / total counts to get percentage
```

```
  ftsum <- colSums(ctmat[ftarr,])
  totsum <- colSums(ctmat)

  perc <- ftsum/totsum
  perc <- perc*100

  # If no metadata create metadata
  if (is.null(scs$metadata)){
    scs$metadata <- data.frame(matrix(nrow=length(perc), ncol=0))
    rownames(scs$metadata) <- names(perc)
  }

  # Add "name" column to metatdata and add calc'd percentage
  scs$metadata[name] <- data.frame(perc )

  return(scs)

}

rm_cells = function(scs,to_rm){

  rm_out <- list()

  rm_out$initial_dim <- scs$initial_dim
  rm_out$metadata <- scs$metadata


  # remove cells with IDs in to_rm
  rm_out$counts <- scs$counts[,setdiff(colnames(scs$counts),to_rm)]

  # remove from metadata
  if (!is.null(scs$metadata)){
  rm_out$metadata <- scs$metadata[setdiff(rownames(scs$metadata),to_rm),,drop=FALSE]

  }

  return(rm_out)

}

rm_genes = function(scs,to_rm){

  rm_out <- list()

  rm_out$initial_dim <- scs$initial_dim
  rm_out$metadata <- scs$metadata

  # remove genes in to_rm
  rm_out$counts <- scs$counts[setdiff(rownames(scs$counts),to_rm),]

  return(rm_out)

}
```

```
larva = read_scs('E:\\BOX\\Box\\DataScienceResource\\Upload to GIthub\\R\\GSM2818521_larva_counts_matri
```

```
## Warning in fread(file): Detected 4365 column names but the data has 4366
## columns (i.e. invalid file). Added an extra default column name for the first
## column which is guessed to be row names or an index. Use setnames() afterwards
## if this guess is not correct, or fix the file write command that created the
## file to create a valid file.
```

```
obj = create_scs(larva)
mt = grep("^MT-", rownames(larva), value = TRUE)
obj = qc_scs(scs = obj,
             min_cells = 30,
             min_features = 300,
             mt_genes = mt,
             max_mt = 6)
```

Normalize

```
normalize_scs = function(scs, scale_factor=10000){

  cts <- scs$counts

  # add all gene counts per cell
  totcts <- colSums(cts)

  # normalize
  normcounts <- t(t(cts)/totcts*scale_factor)
  normcounts <- log(normcounts +1 )

  # insert results into 'normalized'
  scs$normalized <- normcounts

  return(scs)

}

obj = normalize_scs(scs = obj, scale_factor = 10000)
str(obj)
```

```
## List of 4
##  $ initial_dim: Named int [1:2] 24105 4365
##   ..- attr(*, "names")= chr [1:2] "gene" "cell"
##  $ metadata   :'data.frame': 4356 obs. of  1 variable:
##   ..$ percent_mt: num [1:4356] 1.342 0.9 0.872 1.603 1.952 ...
##  $ counts     : int [1:11006, 1:4356] 1 0 0 0 0 0 0 0 1 2 ...
##   ..- attr(*, "dimnames")=List of 2
```

```
##   .. ..$ : chr [1:11006] "PTPRO" "EPS8" "TBK1" "GPR19" ...
##   .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAACCT
##  $ normalized : num [1:11006, 1:4356] 1.14 0 0 0 0 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:11006] "PTPRO" "EPS8" "TBK1" "GPR19" ...
##   .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAACCT
```

print variables in obj's metadata

```
get_vars_scs = function(scs) {

  var_names <- names(scs$metadata)

  return(var_names)

}


get_vars_scs(scs = obj)
```

```
## [1] "percent_mt"
```

Run multiple linear regression on each gene's normalized expression measurements on a variables (vars_to_regress) and then saves the residuals.

```
variation_adjust_scs = function(scs,vars_to_regress=NULL){
  tmp <- scs
  ctnorm <- tmp$normalized
  mtpc <- tmp$metadata[vars_to_regress]
  mtpc <- unlist(mtpc)




    for (i in 1:dim(ctnorm)[1]){

    if (is.null(vars_to_regress)){
    # lm for intercept only for each gene
    ct.lm <- lm(ctnorm[i,]~1)
    } else {
    # lm for each gene
    ct.lm <- lm(ctnorm[i,] ~ mtpc)
    }
    # calc residual and replace
    ct.res <- resid(ct.lm)
    ctnorm[i,] <- ct.res
```

```
    if(i %% 1000 == 0 ){
      print(i)
    }
    }
  tmp$corrected <- ctnorm
  return(tmp)
  }



obj = variation_adjust_scs(scs=obj,vars_to_regress = "percent_mt")
```

```
## [1] 1000
## [1] 2000
## [1] 3000
## [1] 4000
## [1] 5000
## [1] 6000
## [1] 7000
## [1] 8000
## [1] 9000
## [1] 10000
## [1] 11000
```

```
str(obj)
```

```
## List of 5
##  $ initial_dim: Named int [1:2] 24105 4365
##   ..- attr(*, "names")= chr [1:2] "gene" "cell"
##  $ metadata   :'data.frame': 4356 obs. of  1 variable:
##   ..$ percent_mt: num [1:4356] 1.342 0.9 0.872 1.603 1.952 ...
##  $ counts     : int [1:11006, 1:4356] 1 0 0 0 0 0 0 0 1 2 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:11006] "PTPRO" "EPS8" "TBK1" "GPR19" ...
##   .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAACCT
##  $ normalized : num [1:11006, 1:4356] 1.14 0 0 0 0 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:11006] "PTPRO" "EPS8" "TBK1" "GPR19" ...
##   .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAACCT
##  $ corrected  : num [1:11006, 1:4356] 0.8744 -0.0688 -0.2499 -0.1013 -0.1915 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:11006] "PTPRO" "EPS8" "TBK1" "GPR19" ...
##   .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAACCT
```

Identify the most variable genes using num of features are input.

```
library(matrixStats)
variable_feat_scs = function(scs, nfeatures = 2000){
```

```
  tmp <- scs$corrected
  tmpvar <- rowVars(tmp)
  tmpdf <- matrix(tmpvar)
  row.names(tmpdf) <- row.names(tmp)
  maxvargene <- tmpdf[order(tmpdf,decreasing = TRUE),,drop=FALSE]
  maxvar_names <- row.names(maxvargene[1:nfeatures,,drop=FALSE])
  return(maxvar_names)
  # return(maxvargene)
}
```

```
var_feat = variable_feat_scs(scs = obj, nfeatures = 1000)
head(var_feat)
```

```
## [1] "TAC3A"    "SST1.1"   "CCKB"     "ADCYAP1B" "HSP70L"   "GOS2"
```

Perform PCA.

```
run_pca = function(scs, features, npcs){

  # get genes in features from normalized count and run pca
  nvar <- features
  cts.pca  <- prcomp(t(scs$normalized[nvar,]), center = TRUE, scale = TRUE, rank=npcs)

  # get first npcs
  scs$PCA <- cts.pca$x

  return(scs)
}

obj = run_pca(obj,var_feat,10)
str(obj)
```

```
## List of 6
##  $ initial_dim: Named int [1:2] 24105 4365
##   ..- attr(*, "names")= chr [1:2] "gene" "cell"
##  $ metadata   :'data.frame': 4356 obs. of  1 variable:
##   ..$ percent_mt: num [1:4356] 1.342 0.9 0.872 1.603 1.952 ...
##  $ counts     : int [1:11006, 1:4356] 1 0 0 0 0 0 0 0 1 2 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:11006] "PTPRO" "EPS8" "TBK1" "GPR19" ...
##   .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAACCT
##  $ normalized : num [1:11006, 1:4356] 1.14 0 0 0 0 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:11006] "PTPRO" "EPS8" "TBK1" "GPR19" ...
##   .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAACCT
##  $ corrected  : num [1:11006, 1:4356] 0.8744 -0.0688 -0.2499 -0.1013 -0.1915 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:11006] "PTPRO" "EPS8" "TBK1" "GPR19" ...
##   .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAACCT
##  $ PCA        : num [1:4356, 1:10] 3.5 -6.37 -9.04 -2.62 1.11 ...
```

```
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAACCT
##   .. ..$ : chr [1:10] "PC1" "PC2" "PC3" "PC4" ...
```

Get shared nearest neighor network using dbscan library using k = 20, number of nearest neighbors and kt =3, threshold for connection.

```
library("dbscan")
```

```
##
## Attaching package: 'dbscan'

## The following object is masked from 'package:stats':
##
##     as.dendrogram
```

```
make_snn = function(scs, k, kt){

  dt <- scs$PCA
  nn <- sNN(dt, k=k, kt=kt)

  scs$snn <- nn
  return(scs)
}


obj = make_snn(obj, k=20, kt=3)

str(obj)
```

```
## List of 7
##  $ initial_dim: Named int [1:2] 24105 4365
##   ..- attr(*, "names")= chr [1:2] "gene" "cell"
##  $ metadata   :'data.frame': 4356 obs. of  1 variable:
##   ..$ percent_mt: num [1:4356] 1.342 0.9 0.872 1.603 1.952 ...
##  $ counts     : int [1:11006, 1:4356] 1 0 0 0 0 0 0 0 1 2 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:11006] "PTPRO" "EPS8" "TBK1" "GPR19" ...
##   .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAACCT
##  $ normalized : num [1:11006, 1:4356] 1.14 0 0 0 0 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:11006] "PTPRO" "EPS8" "TBK1" "GPR19" ...
##   .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAACCT
##  $ corrected  : num [1:11006, 1:4356] 0.8744 -0.0688 -0.2499 -0.1013 -0.1915 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:11006] "PTPRO" "EPS8" "TBK1" "GPR19" ...
##   .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAACCT
##  $ PCA        : num [1:4356, 1:10] 3.5 -6.37 -9.04 -2.62 1.11 ...
```

```
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAACCT
##   .. ..$ : chr [1:10] "PC1" "PC2" "PC3" "PC4" ...
##  $ snn        :List of 8
##   ..$ dist       : num [1:4356, 1:20] 2.83 4.52 3.26 2.43 4.32 ...
##   .. ..- attr(*, "dimnames")=List of 2
##   .. .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAA
##   .. .. ..$ : chr [1:20] "1" "2" "3" "4" ...
##   ..$ id         : int [1:4356, 1:20] 570 3470 966 3065 1456 1220 1833 4053 1582 625 ...
##   .. ..- attr(*, "dimnames")=List of 2
##   .. .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAA
##   .. .. ..$ : chr [1:20] "1" "2" "3" "4" ...
##   ..$ k          : int 20
##   ..$ sort       : logi FALSE
##   ..$ metric     : chr "euclidean"
##   ..$ shared     : int [1:4356, 1:20] 11 7 9 14 12 10 10 14 12 17 ...
##   ..$ sort_shared: logi TRUE
##   ..$ kt         : num 3
##   ..- attr(*, "class")= chr [1:3] "sNN" "kNN" "NN"
```

Identify clusters using cluster_louvain in igraph library.

```
library("igraph")
```

```
##
## Attaching package: 'igraph'

## The following objects are masked from 'package:stats':
##
##     decompose, spectrum

## The following object is masked from 'package:base':
##
##     union
```

```
find_clusters = function(scs){
  tmp <- scs$snn
  adj <- adjacencylist(tmp)
  g <- graph_from_adj_list(adj, mode = "all", duplicate = FALSE)
  clstr <- cluster_louvain(g)

  scs$metadata$cluster <- as.factor(membership(clstr))
  return(scs)
}

obj = find_clusters(obj)
head(obj$metadata)
```

```
##                            percent_mt cluster
```

```
## larvalR2_AAACCTGAGACAGAGA-1  1.3421389       1
## larvalR2_AAACCTGAGACTTTCG-1  0.9004953       2
## larvalR2_AAACCTGAGAGTAAGG-1  0.8724100       3
## larvalR2_AAACCTGAGAGTCGGT-1  1.6032064       4
## larvalR2_AAACCTGAGGATGGAA-1  1.9515080       5
## larvalR2_AAACCTGAGTAGATGT-1  1.3815789       6
```

Run UMAP using previous results PCA + Clustering

```
library('umap')
run_umap = function(scs){

  res = umap(scs$PCA)
  scs$UMAP = res$layout
  return(scs)

}

obj = run_umap(obj)
str(obj)
```

```
## List of 8
##  $ initial_dim: Named int [1:2] 24105 4365
##   ..- attr(*, "names")= chr [1:2] "gene" "cell"
##  $ metadata   :'data.frame': 4356 obs. of  2 variables:
##   ..$ percent_mt: num [1:4356] 1.342 0.9 0.872 1.603 1.952 ...
##   ..$ cluster   : Factor w/ 17 levels "1","2","3","4",..: 1 2 3 4 5 6 7 6 6 5 ...
##  $ counts     : int [1:11006, 1:4356] 1 0 0 0 0 0 0 0 1 2 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:11006] "PTPRO" "EPS8" "TBK1" "GPR19" ...
##   .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAACCTG
##  $ normalized : num [1:11006, 1:4356] 1.14 0 0 0 0 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:11006] "PTPRO" "EPS8" "TBK1" "GPR19" ...
##   .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAACCTG
##  $ corrected  : num [1:11006, 1:4356] 0.8744 -0.0688 -0.2499 -0.1013 -0.1915 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:11006] "PTPRO" "EPS8" "TBK1" "GPR19" ...
##   .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAACCTG
##  $ PCA        : num [1:4356, 1:10] 3.5 -6.37 -9.04 -2.62 1.11 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAACCTG
##   .. ..$ : chr [1:10] "PC1" "PC2" "PC3" "PC4" ...
##  $ snn        :List of 8
##   ..$ dist       : num [1:4356, 1:20] 2.83 4.52 3.26 2.43 4.32 ...
##   .. ..- attr(*, "dimnames")=List of 2
##   .. .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAAC
##   .. .. ..$ : chr [1:20] "1" "2" "3" "4" ...
##   ..$ id         : int [1:4356, 1:20] 570 3470 966 3065 1456 1220 1833 4053 1582 625 ...
##   .. ..- attr(*, "dimnames")=List of 2
##   .. .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAAC
```

```
##   .. .. ..$ : chr [1:20] "1" "2" "3" "4" ...
##   ..$ k           : int 20
##   ..$ sort        : logi FALSE
##   ..$ metric      : chr "euclidean"
##   ..$ shared      : int [1:4356, 1:20] 11 7 9 14 12 10 10 14 12 17 ...
##   ..$ sort_shared: logi TRUE
##   ..$ kt          : num 3
##   ..- attr(*, "class")= chr [1:3] "sNN" "kNN" "NN"
##  $ UMAP       : num [1:4356, 1:2] 8.86 -4.46 -6.97 -2.17 3.15 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:4356] "larvalR2_AAACCTGAGACAGAGA-1" "larvalR2_AAACCTGAGACTTTCG-1" "larvalR2_AAACCT
##   .. ..$ : NULL
```

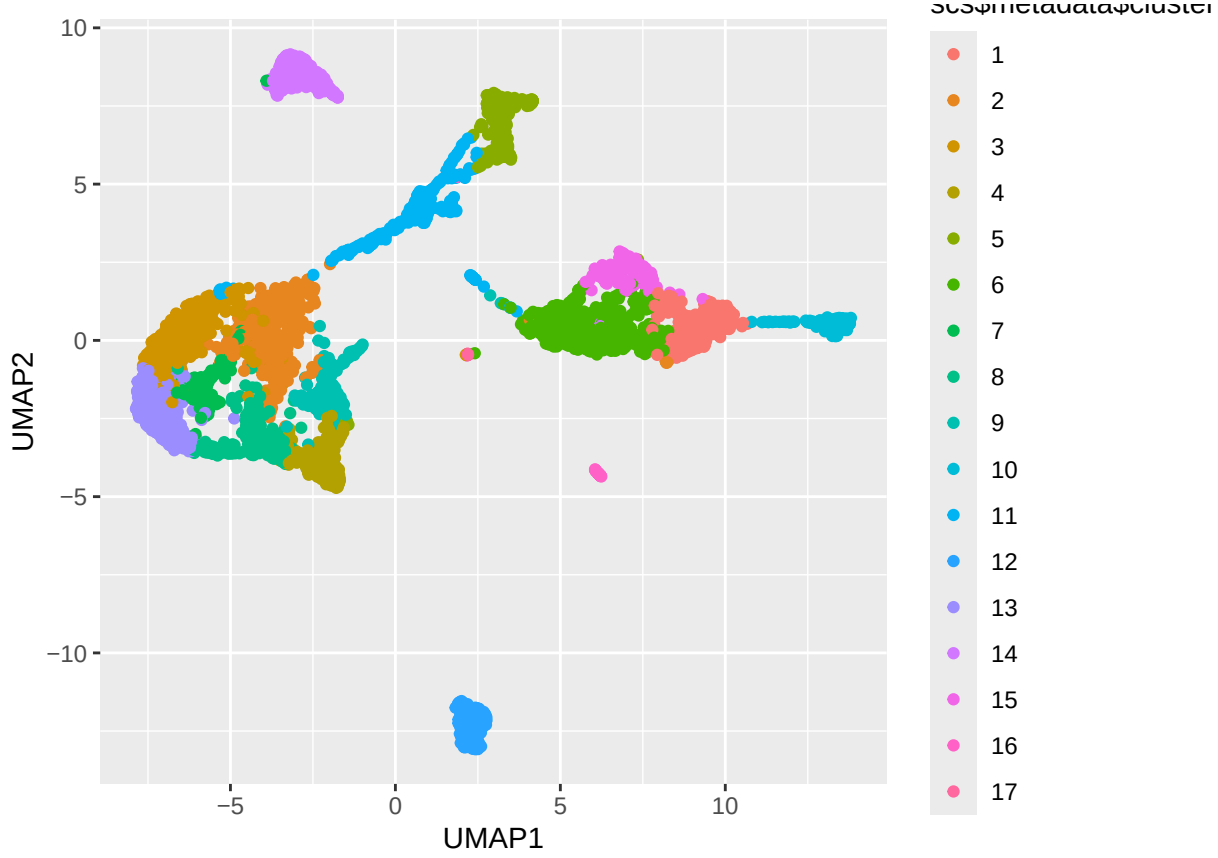
Visualize UMAP

```
library(ggplot2)
viz_scs = function(scs, feature){
  tmp <- scs[[feature]]
  tmp <- as.data.frame(tmp)


ggplot(
  tmp,
  aes(x=V1,
      y=V2,
      color = scs$metadata$cluster )
) +
  geom_point() +
  xlab('UMAP1') + ylab('UMAP2')

}

viz_scs(obj,'UMAP')
```




```
# set.seed(2)
# obj = run_umap(obj)
# viz_scs(obj,'UMAP')
```